

Homework 4

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Problem 7

Problem 7.1. Let G be a finite abelian group, and n coprime to $|G|$. Prove that $a \mapsto a^n$ is an automorphism of G .

Lemma 7.1. Let $\varphi : G \rightarrow H$ be a group homomorphism. Then φ is injective if and only if $\ker \varphi \cong 0$.

Proof. Let $\ker \varphi \cong 0$, and let $g, h \in G$ with $\varphi(g) = \varphi(h)$. It follows that $\varphi(gh^{-1}) = \varphi(e)$, and thus $gh^{-1} \in \ker \varphi$. By assumption, $gh^{-1} = e$, and thus $g = h$, proving injectivity. For the converse, we simply note that $e \in \ker \varphi$ for all φ , and the statement is a triviality since $g \neq e$ implies that $\varphi(g) \neq \varphi(e)$. ■

Proof of problem 7.1. The homomorphism condition is simple. For $g, h \in G$, we have

$$(gh)^n = (gh)(gh) \cdots (gh) = g^n h^n,$$

where the final step follows by commutativity of $\cdot$ in abelian groups. To prove injectivity, it suffices by lemma 7.1 to show the kernel is trivial. Of course, $g^n = e$ if and only if the order of g divides n (the only if direction follows by part of a proof done in the previous homework). However, by Lagrange, the order of g divides the order of G , and since $\gcd(|G|, n) = 1$, we see that $g^n = e$ if and only if $|g| = 1$. It follows that $g = e$, giving us that $\ker(-)^n \cong 0$. Since functions between finite sets of the same cardinality are injective if and only if they are surjective, we have that $(-)^n$ is surjective as well by finiteness of G , showing that it is an automorphism. ■

Problem 8

Problem 8.1. Show that if $G/Z(G)$ is cyclic, then G is abelian.

Proof. If $G/Z(G)$ is cyclic, then it has a generator $xZ(G)$. For all $g, h \in G$, there are thus integers m, n such that $x^n Z(G) = gZ(G)$ and $x^m Z(G) = hZ(G)$. We then have that there is some $z_1, z_2, z_3, z_4 \in Z(G)$ with $x^m z_1 = g z_3$ and $x^n z_2 = h z_4$. We thus have $x^m z_1 z_3^{-1} = g$, and likewise for h . Since $Z(G)$ is a group, we have that $z_1 z_3^{-1} \in Z(G)$. Since elements of the center commute with all elements of G , and since powers of x commute with themselves,

$$gh = x^m z_1 z_3^{-1} x^n z_2 z_4^{-1} = x^n z_2 z_4^{-1} x^m z_1 z_3^{-1} = hg.$$

Therefore, G is abelian. ■

Problem 9

Problem 9.1. Let p be a prime. Show that if G is a finite group of order p^2 , then G is abelian and isomorphic to $\mathbb{Z}/p^2\mathbb{Z}$ or $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, without using the fundamental theorem of finitely generated abelian groups.

Proof. First we show G is abelian. By Lagrange, any subgroup $H \subseteq G$ must have its order divide that of G . Since the only divisors of p^2 are $1, p, p^2$, the order of any subgroup H must have that order.

1. If $|H| = 1$, then $G/H \cong G$.
2. If $|H| = p$, then G/H has order p , and is thus cyclic since all groups of prime order are cyclic.
3. If $|H| = p^2$, then $H = G$, and thus $G/H \cong 0$.

By problem 8.1, it thus suffices to show that $Z(G)$ has more than one element, since it must then have order either p or p^2 . By the class formula,

$$p^2 = |Z(G)| + \sum_{i=1}^k [G : \text{Stab}_G(a_i)],$$

where $\{a_i\}_{i=1}^k$ is a choice of a unique representative for each nontrivial conjugacy class. Since $|Z(G)| \geq 1$ since $e \in Z(G)$, we see that

$$\sum_{i=1}^k [G : \text{Stab}_G(a_i)] < p^2.$$

Since orbit-stabilizer gives

$$[G : \text{Stab}_G(a_i)] = |O_G(a_i)| = \frac{|G|}{|\text{Stab}_G(a_i)|} \implies |\text{Stab}_G(a_i)| = \frac{|G|}{|O_G(a_i)|} \in \mathbb{N}$$

under the conjugacy action, we see that $[G : \text{Stab}_G(a_i)]$ divides $|G|$. Since the sum over all of these is less than p^2 , we see that in particular, each conjugacy class has order either 1 or p . Thus, each stabilizer has order p or p^2 . It suffices to show all stabilizers are proper subgroups. This follows since we will then have that

$$|Z(G)| + \sum_{i=1}^k [G : \text{Stab}_G(a_i)] = |Z(G)| + \sum_{i=1}^k p = |Z(G)| + kp = p^2 \implies |Z(G)| = p(p - k),$$

and since $p \neq 1$, we see that $p(p - k) \neq 1$.

Indeed, suppose for contradiction that $\text{Stab}_G(a_i)$ has order p^2 . Then $\text{Stab}_G(a_i) = G$, and thus $ga_i g^{-1} = a_i$ for all $g \in G$, so all elements of $g \in G$ commute with a_i . But this is precisely the definition of a_i being in the center, so we have $a_i \in Z(G)$, contradicting the assumption that each a_i represents a nontrivial conjugacy class.

To see that $G \cong \mathbb{Z}/p^2\mathbb{Z}$ or $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, we note by Lagrange that all elements of G have order either p or p^2 . Suppose that there is an element g of order p^2 . Then since $|g| = |G|$, we have $\langle g \rangle = G$, and $G \cong \mathbb{Z}/p^2\mathbb{Z}$. For the converse, suppose there is no element of order p^2 . Then all elements have order p . Choose $g \in G$, and choose $h \in G \setminus \langle g \rangle$. Of course $\langle h \rangle \cong \langle g \rangle \cong \mathbb{Z}/p\mathbb{Z}$, so it suffices to show that $G \cong \langle g \rangle \times \langle h \rangle$. This follows directly by noting that all non-identity elements of either subgroup have order p , and thus generate the entire cyclic group. Suppose $\langle g \rangle \cap \langle h \rangle$ has a non-identity element k in it. Then $\langle k \rangle = \langle h \rangle$ and $\langle k \rangle = \langle g \rangle$, but we then see $\langle h \rangle = \langle g \rangle$. Since $h \notin \langle g \rangle$, we obtain a contradiction, showing that $\langle h \rangle \cap \langle g \rangle \cong 0$.

Since G is abelian, we have that $\langle h \rangle \langle g \rangle = \langle g \rangle \langle h \rangle$, so by a lemma shown on homework 5 which I already submitted, we have that $\langle g \rangle \langle h \rangle$ is a subgroup of G isomorphic to $\langle g \rangle \times \langle h \rangle$. It suffices to show $G \subseteq \langle g \rangle \langle h \rangle$. Indeed, we have

$$p^2 = |\langle g \rangle \times \langle h \rangle| = |\langle g \rangle \langle h \rangle| = |G|.$$

It follows that $\langle g \rangle \langle h \rangle = G$, proving the statement. ■

Problem 10

Problem 10.1. Let G be a finite abelian group and p a prime divisor of $|G|$. Show that G has an element of order p .

Proof. We proceed by induction on the order $n = |G|$ of the group. For the base case $|G| = p$, we see that G is cyclic, and thus admits a generator g of order p . Assuming the inductive hypothesis, let $|G| = n$. Choose $x \in G \setminus \{e\}$, and write $k = |x|$. If $p|k$, then we have $k = mp$ for some $m \in \mathbb{Z}$. Let $y = x^m$, and we have

$$|x^m| = \frac{|x|}{\gcd(|x|, m)} = \frac{k}{\gcd(k, m)} = \frac{mp}{m} = p.$$

Therefore, $|y| = p$, and we have found an element of the required order.

If p does not divide k , then the quotient $G/\langle x \rangle$ has order n/k . Write $n = mp$, and thus

$$\left| \frac{G}{\langle x \rangle} \right| = \frac{mp}{k} = \frac{m}{k}p.$$

Since m/k is necessarily an integer, we see that p divides the order of the quotient. By induction, there is an element $g\langle x \rangle$ of order p . We then have $g^p \in \langle x \rangle$, so $g^p = x^j$ for some j . Let $a = |g|$, and since $\pi : G \rightarrow G/\langle x \rangle$ is a group homomorphism, the order of $g\langle x \rangle = \pi(g)$ must divide the order of g (see homework 2 problem 7.3). We thus have $p|a$, meaning $a = pq$ for some $q \geq 1$. Then

$$|g^q| = \frac{a}{\gcd(a, q)} = \frac{pq}{q} = p.$$

The statement follows. ■

Problem 11

Problem 11.1. Let G be a finite group, and $H \subseteq G$ a subgroup with index p , where p is the smallest prime dividing $|G|$. Prove that H is normal by

1. interpreting the action of G on G/H by left-multiplication as a group homomorphism $\sigma : G \rightarrow S_p$;
2. identifying $G/\ker \sigma$ under the first isomorphism theorem with a subgroup of S_p , and considering the implications concerning the index of $\ker \sigma$ in G ;
3. showing that $\ker \sigma \subseteq H$;
4. concluding that $\ker \sigma = H$ by index considerations.

Proof. (1) By definition, since H has index p , the cardinality of the set G/H is p , and thus $\text{Aut}_{\text{set}}(G/H) \cong S_p$ as groups by definition. To interpret left multiplication as an action $G \rightarrow S_p$, it suffices to show left multiplication is a permutation of the cosets of G/H . We then simply identify g with the permutation given by left-multiplication.

Choose representatives $\{a_i\}_{i=1}^p$ of the distinct cosets a_iH , such that the representative of eH is e . Let $g \in G$, and consider the map $a_iH \mapsto ga_iH$. We would like to show this is a permutation. To show surjectivity, let $a_jH \in G/H$. Then there is some a_i with $a_iH = g^{-1}a_jH$ since $g^{-1}a_j \in G$, so we see that $a_iH \mapsto a_jH$. Since G/H is finite, this shows left multiplication is injective as well, proving that left multiplication is a permutation of G/H . We then identify the action with a map

$\sigma : G \rightarrow S_p$ by mapping g to the permutation of $\{1, \dots, p\}$ on the indices of the a_i s given by left multiplication on G/H .

(2) Identifying along the first isomorphism theorem, we have $G/\ker \sigma \subseteq S_p$. We thus have by Lagrange that $[G : \ker \sigma]$ divides $|S_p| = p!$ and $|G|$. In particular, $[G : \ker \sigma] \leq \gcd(p!, |G|)$.

(3) Let $g \in G$ induce the identity map on G/H under σ . Then in particular we have $gH = H$, which follows if and only if $g \in H$ by problem 1.a. This gives $\ker \sigma \subseteq H$.

(4) Of course, the only possible prime divisors of $p!$ are primes $q \leq p$, and the only possible prime divisors of $|G|$ are primes $q \geq p$. We thus see that the only possible prime divisor of $[G : \ker \sigma]$ is p . Since $[G : \ker \sigma]$ is just a number, it admits a prime decomposition into its prime divisors, which in this case are only p . Thus, $[G : \ker \sigma]$ is a prime power of p . In particular, $p \leq [G : \ker \sigma]$. However, the largest power of p dividing $p!$ is p , so we have that $[G : \ker \sigma] = p$. In particular, $|\ker \sigma| = |H|$. Since these are finite subgroups and we already have $\ker \sigma \subseteq H$, we may conclude that $\ker \sigma = H$. Since kernels are normal, the statement is proven. ■